



Description
No Data

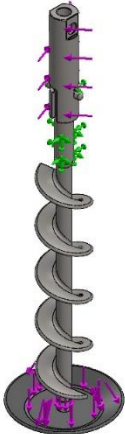
Simulation of auger and shaft and plate

Date: Tuesday, January 6, 2026
Designer: Solidworks
Study name: auger and shaft and plate analysis
Analysis type: Static
ENG name: Hassan Moqbel

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Assumptions

Model Information



Model name: auger and shaft and plate
Current Configuration: Default

Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Combine1 	Solid Body	Mass:1.96918 kg Volume:0.000246147 m^3 Density:8,000.01 kg/m^3 Weight:19.2979 N	C:\Users\user1\Desktop\VFFS\analysis\solidworks analysis\auger and shaft and pannel\auger and shaft and plate.SLDPRT Jan 6 21:40:52 2026

Study Properties

Study name	auger and shaft and plate analysis
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\user1\Desktop\VFFS\analysis\solidworks analysis\auger and shaft and pannel)

Units

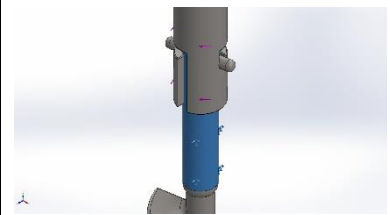
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²



Material Properties

Model Reference	Properties	Components
	Name: AISI 304 Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 2.06807e+08 N/m ² Tensile strength: 5.17017e+08 N/m ² Elastic modulus: 1.9e+11 N/m ² Poisson's ratio: 0.29 Mass density: 8,000 kg/m ³ Shear modulus: 7.5e+10 N/m ² Thermal expansion coefficient: 1.8e-05 /Kelvin	SolidBody 1(Combine1)(auger and shaft and plate)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry

Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	0.999783	14.139	-0.000153375	14.1743
Reaction Moment(N.m)	0	0	0	0

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<LoadName/>	<Image_Load/>	<Details_Load/>



Connector Definitions

No Data

Interaction Information

No Data

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	6.58288 mm
Minimum element size	0.329144 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	570587
Total Elements	371941
Maximum Aspect Ratio	69.405
% of elements with Aspect Ratio < 3	99.2
Percentage of elements with Aspect Ratio > 10	0.0194
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:33
Computer name:	

Sensor Details

No Data



Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.999783	14.139	-0.000153375	14.1743

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.0879457	0.0406183	0.0510134	0.109484

Free body moments

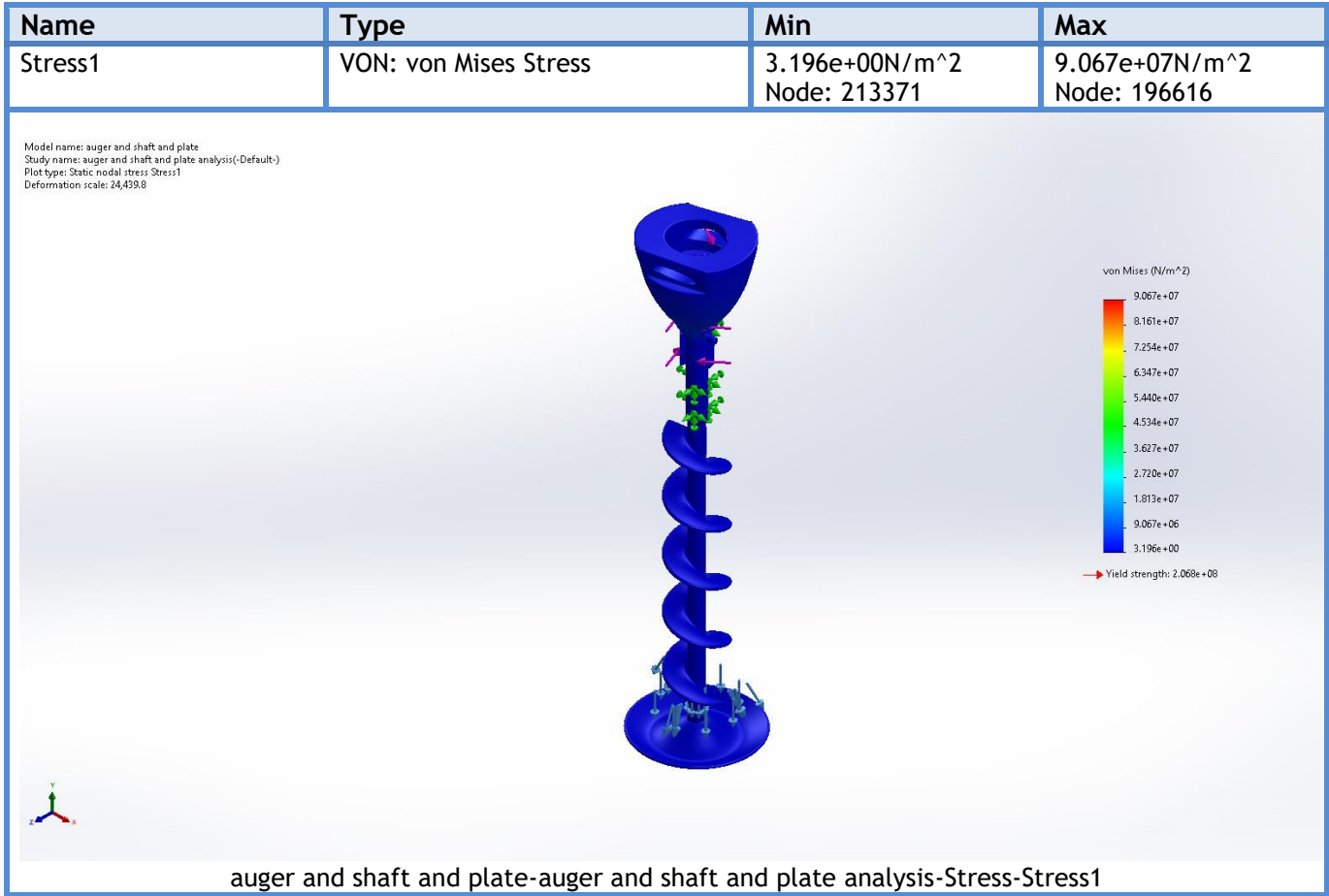
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

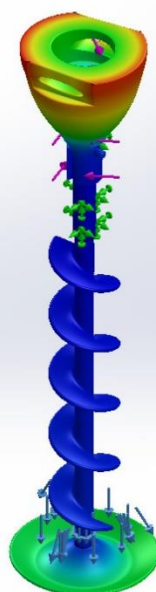


Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 136	2.137e-03mm Node: 4337

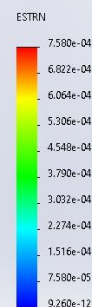
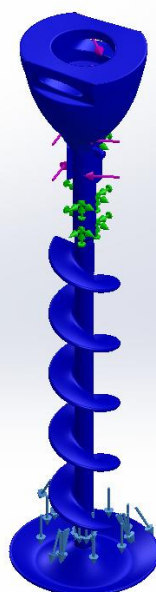
Model name: auger and shaft and plate
Study name: auger and shaft and plate analysis(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 24,439.8



auger and shaft and plate-auger and shaft and plate analysis-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	9.260e-12 Element: 101310	7.580e-04 Element: 82398

Model name: auger and shaft and plate
Study name: auger and shaft and plate analysis(-Default-)
Plot type: Static strain Strain1
Deformation scale: 24,439.8

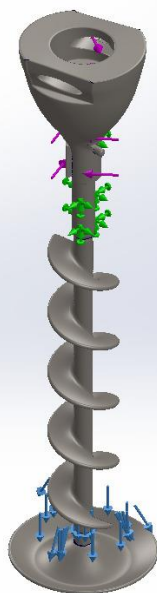


auger and shaft and plate-auger and shaft and plate analysis-Strain-Strain1

Name	Type
Displacement1{1}	Deformed shape



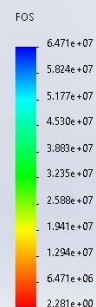
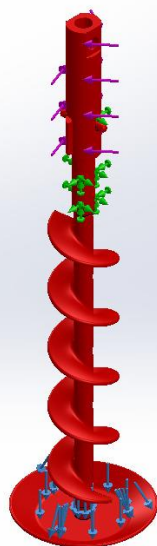
Model name: auger and shaft and plate
 Study name: auger and shaft and plate analysis(-Default-)
 Plot type: Deformed shape Displacement1[1]
 Deformation scale: 24,439.8



auger and shaft and plate-auger and shaft and plate analysis-Displacement-Displacement1{1}

Name	Type	Min	Max
Factor of Safety1	Max von Mises Stress	2.281e+00 Node: 196616	6.471e+07 Node: 213371

Model name: auger and shaft and plate
 Study name: auger and shaft and plate analysis(-Default-)
 Plot type: Factor of Safety Factor of Safety1
 Criterion : Max von Mises Stress
 Factor of safety distribution: Min FOS = 2.3



auger and shaft and plate-auger and shaft and plate analysis-Factor of Safety-Factor of Safety1



Conclusion

